Problem solving with transformations

Task A: Inverse problems

1) Shape M is translated by $\begin{pmatrix} 3 \\ 2 \end{pmatrix}$ to create shape N. What vector would translate shape N onto shape M?

$\begin{pmatrix} -3 \\ -2 \end{pmatrix}$

2) Shape P is reflected in the x-axis to create shape Q. What transformation would map shape Q back on to shape P?

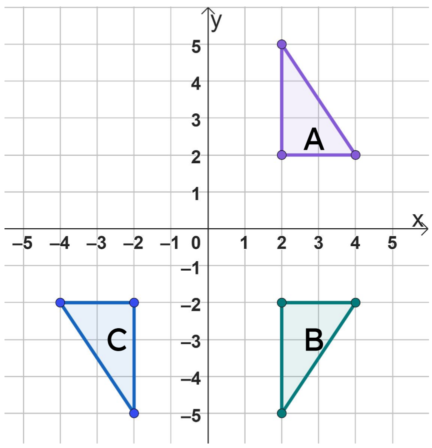
Reflection in the x-axis.
(Depending on shape P it could be a translation or rotation as well).

3) Shape R is enlarged by a scale factor of 4 from the origin. What enlargement scale factor would be needed to map back to shape R? $\frac{1}{4}$

4) Shape A is reflected in the x-axis to become shape B. Shape B is then reflected in the y-axis to become shape C.

Fully describe a single transformation that can map shape C onto shape A.

Rotate shape C 180° about the origin.



Task B: Perimeter and area

1) A shape has a perimeter of 10 cm.
   a) If it is translated its perimeter would be *10* cm.
   b) If it is rotated its perimeter would be *10* cm.
   c) If it is reflected its perimeter would be *10* cm.
   d) If it is enlarged by scale factor 0.5, its perimeter would be *5* cm.

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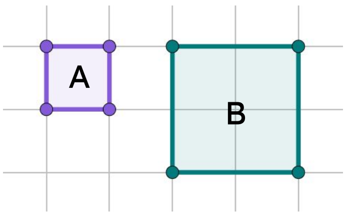
2) Work out the perimeter and area of the object and image in the following enlargements. Use 1 square = 1 cm.

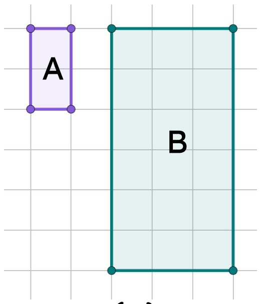
a)



Perimeter (A)= 4 cm
Area (A) = 1 cm^2

Perimeter (B) = 8 cm
Area (B) = 4 cm^2

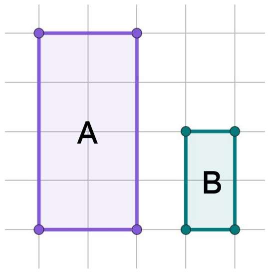
b)



Perimeter (A)= 6 cm
Area (A) = 2 cm^2

Perimeter (B) = 18 cm
Area (B) = 18 cm^2

c)



Perimeter (A)= 12 cm
Area (A) = 8 cm^2

Perimeter (B) = 6 cm
Area (B) = 2 cm^2

Task C: Coordinate problems

1) The following points have been translated by $\begin{pmatrix} -5 \\ 3 \end{pmatrix}$
Write the coordinate of the image.

a) (a, b) $(a-5, b+3)$

b) $(a+1, b+3)$ $(a-4, b+6)$

c) $(a-3, b+3)$ $(a-8, b+6)$

d) $(2a, 7-b)$ $(2a-5, 10-b)$

e) $(4+a, b+2)$ $(a-1, b+5)$

2) If these are the coordinates for the vertices, what transformation has occurred?

a) Object: A(1,4) B(1,1) C(2,1)
Image: A′(4,4) B′(4,1) C′(5,1)

Translated by $\begin{pmatrix} 3 \\ 0 \end{pmatrix}$

b) Object: A(1,4) B(1,1) C(2,1)
Image: A′(1,4) B′(1,7) C′(2,7)

Reflected in a horizontal line.